

one term per choice of the two free columns $U = \{j_1 < j_2\}$

and every choice falls into one of three types

	β_1	β_2	β_3	β_4	β_5	β_6
1	$x_1^{\beta_1}$	$x_1^{\beta_2}$	$x_1^{\beta_3}$	$x_1^{\beta_4}$	$x_1^{\beta_5}$	$x_1^{\beta_6}$
ζ	$x_2^{\beta_1}$	$x_2^{\beta_2}$	$x_2^{\beta_3}$	$x_2^{\beta_4}$	$x_2^{\beta_5}$	$x_2^{\beta_6}$
ζ^2	$x_3^{\beta_1}$	$x_3^{\beta_2}$	$x_3^{\beta_3}$	$x_3^{\beta_4}$	$x_3^{\beta_5}$	$x_3^{\beta_6}$
ζ^3	$x_4^{\beta_1}$	$x_4^{\beta_2}$	$x_4^{\beta_3}$	$x_4^{\beta_4}$	$x_4^{\beta_5}$	$x_4^{\beta_6}$
z	$x_5^{\beta_1}$	$x_5^{\beta_2}$	$x_5^{\beta_3}$	$x_5^{\beta_4}$	$x_5^{\beta_5}$	$x_5^{\beta_6}$
z^{-1}	$x_6^{\beta_1}$	$x_6^{\beta_2}$	$x_6^{\beta_3}$	$x_6^{\beta_4}$	$x_6^{\beta_5}$	$x_6^{\beta_6}$

$$M_S = (-1)^{\text{inv}(b_S)} V$$

Lemma 4.1: zero unless the t residues on S are distinct

$f(\beta_{j_1} - \beta_{j_2})$
the free 2×2 block



a residue class is empty
profile (3, 2, 1, 0)
0 terms



two classes of size two
profile (2, 2, 1, 1)
4 terms



one class of size three
profile (3, 1, 1, 1)
3 terms

● residue 0 ● residue 1 ● residue 2 ● residue 3